324 BEST KNOWN SQUARE PACKINGS

★ Including new results (October 8, 2026)

github.com/jlevy/squares

n = 1..324; 18 square-bound rows; 52,650 unit squares

O proved optimal (77)
= exact value known (287)
≈ only known numerically (37)
R rigid (established here) (20)
R annotated rigid by the catalogue (2)
★ recent result, since Aug 2026 (297)
🟩🟨🟪🟦 colors indicate distinct tilt angles
4 3 2 1 0 shade indicates number of full-side contacts

s(n) is the side of the smallest square holding n unit squares; deg is the algebraic degree of that side length
Citations for all results are available in the Squares Project: github.com/jlevy/squares
Diagram by Joshua Levy with assistance from Claude and Codex
v0.5.0-4550ad

1	O = R	s(1) = 1		
2	O =	s(2) = 2		
3	O =	s(3) = 2		
4	O = R	s(4) = 2		
5	O = R	s(5) = 2.707107	deg 2	
6	O =	s(6) = 3		
7	★ O =	s(7) = 3		
8	★ O =	s(8) = 3		
9	O = R	s(9) = 3		
10	O =	s(10) = 3.707107	deg 2	
11	★ O = R	s(11) = 3.877084	deg 8	
12	★ =	s(12) ≤ 4		s(12) ≥ 3.971500
13	O =	s(13) = 4		
14	★ O =	s(14) = 4		
15	★ O =	s(15) = 4		
16	O = R	s(16) = 4		
17	★ =	s(17) ≤ 4.675531	deg 18	s(17) ≥ 4.660442
18	★ =	s(18) ≤ 4.822876	deg 2	s(18) ≥ 4.705000
19	★ =	s(19) ≤ 4.885619	deg 2	s(19) ≥ 4.825000
20	★ =	s(20) ≤ 5		s(20) ≥ 4.905000
21	★ O =	s(21) = 5		
22	O =	s(22) = 5		
23	★ O =	s(23) = 5		
24	★ O =	s(24) = 5		
25	O = R	s(25) = 5		
26	★ =	s(26) ≤ 5.621321	deg 2	s(26) ≥ 5.545000
27	★ =	s(27) ≤ 5.707107	deg 2	s(27) ≥ 5.643500
28	★ = R	s(28) ≤ 5.824445	deg 6	s(28) ≥ 5.735000
29	★ ≈	s(29) ≤ 5.933834		s(29) ≥ 5.810000
30	★ =	s(30) ≤ 6		s(30) ≥ 5.883500
31	★ =	s(31) ≤ 6		s(31) ≥ 5.952500
32	★ O =	s(32) = 6		
33	O =	s(33) = 6		
34	★ O =	s(34) = 6		
35	★ O =	s(35) = 6		
36	O = R	s(36) = 6		
37	★ =	s(37) ≤ 6.598620	deg 8	s(37) ≥ 6.440000
38	★ =	s(38) ≤ 6.707107	deg 2	s(38) ≥ 6.545000
39	★ =	s(39) ≤ 6.810723	deg 5	s(39) ≥ 6.650000
40	★ = R	s(40) ≤ 6.828428	deg 2	s(40) ≥ 6.700000
41	★ =	s(41) ≤ 6.926694	deg 42	s(41) ≥ 6.775000
42	★ =	s(42) ≤ 7		s(42) ≥ 6.847500
43	★ =	s(43) ≤ 7		s(43) ≥ 6.907500
44	★ =	s(44) ≤ 7		s(44) ≥ 6.972500
45	★ O =	s(45) = 7		
46	O =	s(46) = 7		
47	★ O =	s(47) = 7		
48	★ O =	s(48) = 7		
49	O = R	s(49) = 7		
50	★ =	s(50) ≤ 7.571429		s(50) ≥ 7.400000
51	★ =	s(51) ≤ 7.700800	deg 12	s(51) ≥ 7.470000
52	★ =	s(52) ≤ 7.707107	deg 2	s(52) ≥ 7.550000
53	★ =	s(53) ≤ 7.822876	deg 2	s(53) ≥ 7.627500
54	★ =	s(54) ≤ 7.846668	deg 4	s(54) ≥ 7.685000
55	★ ≈	s(55) ≤ 7.945772		s(55) ≥ 7.728000
56	★ =	s(56) ≤ 8		s(56) ≥ 7.802500
57	★ =	s(57) ≤ 8		s(57) ≥ 7.872500
58	★ =	s(58) ≤ 8		s(58) ≥ 7.935000
59	★ O =	s(59) = 8		
60	★ O =	s(60) = 8		
61	★ O =	s(61) = 8		
62	★ O =	s(62) = 8		
63	★ O =	s(63) = 8		
64	O = R	s(64) = 8		
65	★ =	s(65) ≤ 8.535534	deg 2	s(65) ≥ 8.350000
66	★ =	s(66) ≤ 8.656855	deg 2	s(66) ≥ 8.430000
67	★ =	s(67) ≤ 8.707107	deg 2	s(67) ≥ 8.480000
68	★ ≈	s(68) ≤ 8.798796		s(68) ≥ 8.510000
69	★ =	s(69) ≤ 8.827195	deg 38	s(69) ≥ 8.620000
70	★ =	s(70) ≤ 8.881667	deg 4	s(70) ≥ 8.657500
71	★ ≈	s(71) ≤ 8.944072		s(71) ≥ 8.721000
72	★ =	s(72) ≤ 9		s(72) ≥ 8.760000
73	★ =	s(73) ≤ 9		s(73) ≥ 8.813000
74	★ =	s(74) ≤ 9		s(74) ≥ 8.867500
75	★ =	s(75) ≤ 9		s(75) ≥ 8.940000
76	★ =	s(76) ≤ 9		s(76) ≥ 8.965000
77	★ O =	s(77) = 9		
78	★ O =	s(78) = 9		
79	★ O =	s(79) = 9		
80	★ O =	s(80) = 9		
81	O = R	s(81) = 9		
82	★ =	s(82) ≤ 9.535534	deg 2	s(82) ≥ 9.320000
83	★ =	s(83) ≤ 9.634758	deg 672	s(83) ≥ 9.370000
84	★ =	s(84) ≤ 9.707107	deg 2	s(84) ≥ 9.411000
85	★ =	s(85) ≤ 9.742641	deg 2	s(85) ≥ 9.460000
86	★ =	s(86) ≤ 9.822876	deg 2	s(86) ≥ 9.503000
87	★ =	s(87) ≤ 9.838816	deg 41	s(87) ≥ 9.580000
88	★ =	s(88) ≤ 9.882452		s(88) ≥ 9.620000
89	★ =	s(89) ≤ 9.949748	deg 2	s(89) ≥ 9.650000
90	★ =	s(90) ≤ 10		s(90) ≥ 9.730000
91	★ =	s(91) ≤ 10		s(91) ≥ 9.762500
92	★ =	s(92) ≤ 10		s(92) ≥ 9.770000
93	★ =	s(93) ≤ 10		s(93) ≥ 9.880000
94	★ =	s(94) ≤ 10		s(94) ≥ 9.950000
95	★ =	s(95) ≤ 10		s(95) ≥ 9.965000
96	★ =	s(96) ≤ 10		s(96) ≥ 9.970000
97	★ O =	s(97) = 10		
98	★ O =	s(98) = 10		
99	★ O =	s(99) = 10		
100	O = R	s(100) = 10		
101	★ =	s(101) ≤ 10.535534	deg 2	s(101) ≥ 10.280000
102	★ ≈	s(102) ≤ 10.607175		s(102) ≥ 10.280000
103	★ ≈	s(103) ≤ 10.703517		s(103) ≥ 10.280000
104	★ =	s(104) ≤ 10.707107	deg 2	s(104) ≥ 10.280000
105	★ ≈	s(105) ≤ 10.806078		s(105) ≥ 10.280000
106	★ ≈	s(106) ≤ 10.822909		s(106) ≥ 10.310708
107	★ =	s(107) ≤ 10.846668	deg 4	s(107) ≥ 10.361541
108	★ =	s(108) ≤ 10.909941		s(108) ≥ 10.412113
109	★ =	s(109) ≤ 10.949748	deg 2	s(109) ≥ 10.462429
110	★ ≈	s(110) ≤ 10.996784		s(110) ≥ 10.512492
111	★ =	s(111) ≤ 11		s(111) ≥ 10.562305
112	★ =	s(112) ≤ 11		s(112) ≥ 10.611874
113	★ =	s(113) ≤ 11		s(113) ≥ 10.661200
114	★ =	s(114) ≤ 11		s(114) ≥ 10.710288
115	★ =	s(115) ≤ 11		s(115) ≥ 10.759142
116	★ =	s(116) ≤ 11		s(116) ≥ 10.807764
117	★ =	s(117) ≤ 11		s(117) ≥ 10.856157
118	★ O =	s(118) = 11		
119	★ O =	s(119) = 11		
120	★ O =	s(120) = 11		
121	O = R	s(121) = 11		
122	★ =	s(122) ≤ 11.535534	deg 2	s(122) ≥ 11.047511
123	★ =	s(123) ≤ 11.600908		s(123) ≥ 11.094810
124	★ =	s(124) ≤ 11.656855	deg 2	s(124) ≥ 11.141898
125	★ =	s(125) ≤ 11.707107	deg 2	s(125) ≥ 11.188779
126	★ =	s(126) ≤ 11.773304		s(126) ≥ 11.235455
127	★ =	s(127) ≤ 11.822876	deg 2	s(127) ≥ 11.281929
128	★ =	s(128) ≤ 11.825092	deg 40	s(128) ≥ 11.328203
129	★ =	s(129) ≤ 11.879376		s(129) ≥ 11.374281
130	★ =	s(130) ≤ 11.904484		s(130) ≥ 11.420164
131	★ ≈	s(131) ≤ 11.954917		s(131) ≥ 11.465856
132	★ ≈	s(132) ≤ 11.991328		s(132) ≥ 11.511357
133	★ =	s(133) ≤ 12		s(133) ≥ 11.556672
134	★ =	s(134) ≤ 12		s(134) ≥ 11.601801
135	★ =	s(135) ≤ 12		s(135) ≥ 11.646748
136	★ =	s(136) ≤ 12		s(136) ≥ 11.691514
137	★ =	s(137) ≤ 12		s(137) ≥ 11.736102
138	★ =	s(138) ≤ 12		s(138) ≥ 11.780514
139	★ =	s(139) ≤ 12		s(139) ≥ 11.824751
140	★ =	s(140) ≤ 12		s(140) ≥ 11.868817
141	★ O =	s(141) = 12		
142	★ O =	s(142) = 12		
143	★ O =	s(143) = 12		
144	O = R	s(144) = 12		
145	★ =	s(145) ≤ 12.535534	deg 2	s(145) ≥ 12.043396
146	★ =	s(146) ≤ 12.600908	deg 16	s(146) ≥ 12.086630
147	★ =	s(147) ≤ 12.656855	deg 2	s(147) ≥ 12.129703
148	★ =	s(148) ≤ 12.656855	deg 2	s(148) ≥ 12.172617
149	★ =	s(149) ≤ 12.707107	deg 2	s(149) ≥ 12.215374
150	★ =	s(150) ≤ 12.778175	deg 2	s(150) ≥ 12.257976
151	★ =	s(151) ≤ 12.822876	deg 2	s(151) ≥ 12.300423
152	★ ≈	s(152) ≤ 12.830719		s(152) ≥ 12.342719
153	★ =	s(153) ≤ 12.879680		s(153) ≥ 12.384864
154	★ =	s(154) ≤ 12.926563		s(154) ≥ 12.426860
155	★ =	s(155) ≤ 12.952504		s(155) ≥ 12.468709
156	★ ≈	s(156) ≤ 12.982083		s(156) ≥ 12.510412
157	★ =	s(157) ≤ 13		s(157) ≥ 12.551970
158	★ =	s(158) ≤ 13		s(158) ≥ 12.593386
159	★ =	s(159) ≤ 13		s(159) ≥ 12.634661
160	★ =	s(160) ≤ 13		s(160) ≥ 12.675795
161	★ =	s(161) ≤ 13		s(161) ≥ 12.716791
162	★ =	s(162) ≤ 13		s(162) ≥ 12.757650
163	★ =	s(163) ≤ 13		s(163) ≥ 12.798373
164	★ =	s(164) ≤ 13		s(164) ≥ 12.838962
165	★ =	s(165) ≤ 13		s(165) ≥ 12.879418
166	★ O =	s(166) = 13		
167	★ O =	s(167) = 13		
168	★ O =	s(168) = 13		
169	O = R	s(169) = 13		
170	★ =	s(170) ≤ 13.535534	deg 2	s(170) ≥ 13.039936
171	★ =	s(171) ≤ 13.571429		s(171) ≥ 13.079745
172	★ ≈	s(172) ≤ 13.618989		s(172) ≥ 13.119429
173	★ =	s(173) ≤ 13.656855	deg 2	s(173) ≥ 13.158988
174	★ =	s(174) ≤ 13.707107	deg 2	s(174) ≥ 13.198425
175	★ =	s(175) ≤ 13.778175	deg 2	s(175) ≥ 13.237739
176	★ =	s(176) ≤ 13.822876	deg 2	s(176) ≥ 13.276932
177	★ ≈	s(177) ≤ 13.822980		s(177) ≥ 13.316005
178	★ =	s(178) ≤ 13.846668	deg 4	s(178) ≥ 13.354960
179	★ =	s(179) ≤ 13.883796		s(179) ≥ 13.393796
180	★ =	s(180) ≤ 13.917654		s(180) ≥ 13.432517
181	★ ≈	s(181) ≤ 13.953749		s(181) ≥ 13.471121
182	★ ≈	s(182) ≤ 13.974091		s(182) ≥ 13.509611
183	★ =	s(183) ≤ 14		s(183) ≥ 13.547988
184	★ =	s(184) ≤ 14		s(184) ≥ 13.586252
185	★ =	s(185) ≤ 14		s(185) ≥ 13.624404
186	★ =	s(186) ≤ 14		s(186) ≥ 13.662446
187	★ =	s(187) ≤ 14		s(187) ≥ 13.700378
188	★ =	s(188) ≤ 14		s(188) ≥ 13.738202
189	★ =	s(189) ≤ 14		s(189) ≥ 13.775918
190	★ =	s(190) ≤ 14		s(190) ≥ 13.813526
191	★ =	s(191) ≤ 14		s(191) ≥ 13.851029
192	★ =	s(192) ≤ 14		s(192) ≥ 13.888427
193	★ O =	s(193) = 14		
194	★ O =	s(194) = 14		
195	★ O =	s(195) = 14		
196	O = R	s(196) = 14		
197	★ =	s(197) ≤ 14.535534	deg 2	s(197) ≥ 14.036986
198	★ =	s(198) ≤ 14.571429		s(198) ≥ 14.073871
199	★ ≈	s(199) ≤ 14.617573		s(199) ≥ 14.110657
200	★ =	s(200) ≤ 14.656855	deg 2	s(200) ≥ 14.147344
201	★ =	s(201) ≤ 14.707107	deg 2	s(201) ≥ 14.183932
202	★ =	s(202) ≤ 14.727923	deg 2	s(202) ≥ 14.220422
203	★ =	s(203) ≤ 14.778175	deg 2	s(203) ≥ 14.256816
204	★ =	s(204) ≤ 14.822876	deg 2	s(204) ≥ 14.293114
205	★ =	s(205) ≤ 14.824452	deg 40	s(205) ≥ 14.329316
206	★ ≈	s(206) ≤ 14.860159		s(206) ≥ 14.365424
207	★ =	s(207) ≤ 14.887993		s(207) ≥ 14.401438
208	★ =	s(208) ≤ 14.924519		s(208) ≥ 14.437359
209	★ =	s(209) ≤ 14.949618		s(209) ≥ 14.473188
210	★ ≈	s(210) ≤ 14.973002		s(210) ≥ 14.508925
211	★ ≈	s(211) ≤ 14.997961		s(211) ≥ 14.544571
212	★ =	s(212) ≤ 15		s(212) ≥ 14.580127
213	★ =	s(213) ≤ 15		s(213) ≥ 14.615594
214	★ =	s(214) ≤ 15		s(214) ≥ 14.650971
215	★ =	s(215) ≤ 15		s(215) ≥ 14.686260
216	★ =	s(216) ≤ 15		s(216) ≥ 14.721462
217	★ =	s(217) ≤ 15		s(217) ≥ 14.756577
218	★ =	s(218) ≤ 15		s(218) ≥ 14.791605
219	★ =	s(219) ≤ 15		s(219) ≥ 14.826548
220	★ =	s(220) ≤ 15		s(220) ≥ 14.861406
221	★ =	s(221) ≤ 15		s(221) ≥ 14.896180
222	★ O =	s(222) = 15		
223	★ O =	s(223) = 15		
224	★ O =	s(224) = 15		
225	O = R	s(225) = 15		
226	★ =	s(226) ≤ 15.535534	deg 2	s(226) ≥ 15.034441
227	★ =	s(227) ≤ 15.571068	deg 2	s(227) ≥ 15.068802
228	★ ≈	s(228) ≤ 15.604603		s(228) ≥ 15.103081
229	★ =	s(229) ≤ 15.656855	deg 2	s(229) ≥ 15.137281
230	★ =	s(230) ≤ 15.682927		s(230) ≥ 15.171400
231	★ =	s(231) ≤ 15.707107	deg 2	s(231) ≥ 15.205441
232	★ =	s(232) ≤ 15.778175	deg 2	s(232) ≥ 15.239402
233	★ =	s(233) ≤ 15.778175	deg 2	s(233) ≥ 15.273286
234	★ =	s(234) ≤ 15.822876	deg 2	s(234) ≥ 15.307092
235	★ =	s(235) ≤ 15.826606	deg 83	s(235) ≥ 15.340822
236	★ =	s(236) ≤ 15.867801		s(236) ≥ 15.374474
237	★ =	s(237) ≤ 15.903677		s(237) ≥ 15.408051
238	★ =	s(238) ≤ 15.926147		s(238) ≥ 15.441552
239	★ =	s(239) ≤ 15.949314		s(239) ≥ 15.474979
240	★ ≈	s(240) ≤ 15.969686		s(240) ≥ 15.508331
241	★ ≈	s(241) ≤ 15.988133		s(241) ≥ 15.541608
242	★ =	s(242) ≤ 16		s(242) ≥ 15.574813
243	★ =	s(243) ≤ 16		s(243) ≥ 15.607944
244	★ =	s(244) ≤ 16		s(244) ≥ 15.641003
245	★ =	s(245) ≤ 16		s(245) ≥ 15.673990
246	★ =	s(246) ≤ 16		s(246) ≥ 15.706906
247	★ =	s(247) ≤ 16		s(247) ≥ 15.739750
248	★ =	s(248) ≤ 16		s(248) ≥ 15.772524
249	★ =	s(249) ≤ 16		s(249) ≥ 15.805227
250	★ =	s(250) ≤ 16		s(250) ≥ 15.837861
251	★ =	s(251) ≤ 16		s(251) ≥ 15.870426
252	★ =	s(252) ≤ 16		s(252) ≥ 15.902921
253	★ O =	s(253) = 16		
254	★ O =	s(254) = 16		
255	★ O =	s(255) = 16		
256	O = R	s(256) = 16		
257	★ =	s(257) ≤ 16.535534	deg 2	s(257) ≥ 16.032224
258	★ =	s(258) ≤ 16.563449		s(258) ≥ 16.064382
259	★ ≈	s(259) ≤ 16.602569		s(259) ≥ 16.096473
260	★ =	s(260) ≤ 16.656855	deg 2	s(260) ≥ 16.128499
261	★ =	s(261) ≤ 16.682927		s(261) ≥ 16.160459
262	★ =	s(262) ≤ 16.707107	deg 2	s(262) ≥ 16.192354
263	★ =	s(263) ≤ 16.740420		s(263) ≥ 16.224185
264	★ =	s(264) ≤ 16.778175	deg 2	s(264) ≥ 16.255951
265	★ =	s(265) ≤ 16.778175	deg 2	s(265) ≥ 16.287653
266	★ =	s(266) ≤ 16.823066	deg 32	s(266) ≥ 16.319292
267	★ =	s(267) ≤ 16.846668	deg 4	s(267) ≥ 16.350867
268	★ ≈	s(268) ≤ 16.878815		s(268) ≥ 16.382380
269	★ ≈	s(269) ≤ 16.905968		s(269) ≥ 16.413830
270	★ ≈	s(270) ≤ 16.937811		s(270) ≥ 16.445218
271	★ ≈	s(271) ≤ 16.950821		s(271) ≥ 16.476545
272	★ ≈	s(272) ≤ 16.968166		s(272) ≥ 16.507810
273	★ ≈	s(273) ≤ 16.983926		s(273) ≥ 16.539014
274	★ =	s(274) ≤ 17		s(274) ≥ 16.570158
275	★ =	s(275) ≤ 17		s(275) ≥ 16.601242
276	★ =	s(276) ≤ 17		s(276) ≥ 16.632265
277	★ =	s(277) ≤ 17		s(277) ≥ 16.663229
278	★ =	s(278) ≤ 17		s(278) ≥ 16.694134
279	★ =	s(279) ≤ 17		s(279) ≥ 16.724980
280	★ =	s(280) ≤ 17		s(280) ≥ 16.755768
281	★ =	s(281) ≤ 17		s(281) ≥ 16.786497
282	★ =	s(282) ≤ 17		s(282) ≥ 16.817168
283	★ =	s(283) ≤ 17		s(283) ≥ 16.847782
284	★ =	s(284) ≤ 17		s(284) ≥ 16.878339
285	★ =	s(285) ≤ 17		s(285) ≥ 16.908839
286	★ O =	s(286) = 17		
287	★ O =	s(287) = 17		
288	★ O =	s(288) = 17		
289	O = R	s(289) = 17		
290	★ =	s(290) ≤ 17.535534	deg 2	s(290) ≥ 17.030275
291	★ =	s(291) ≤ 17.535534	deg 2	s(291) ≥ 17.060495
292	★ ≈	s(292) ≤ 17.597250		s(292) ≥ 17.090660
293	★ =	s(293) ≤ 17.634147		s(293) ≥ 17.120770
294	★ =	s(294) ≤ 17.656855	deg 2	s(294) ≥ 17.150825
295	★ =	s(295) ≤ 17.707107	deg 2	s(295) ≥ 17.180827
296	★ =	s(296) ≤ 17.707107	deg 2	s(296) ≥ 17.210774
297	★ ≈	s(297) ≤ 17.740418		s(297) ≥ 17.240669
298	★ =	s(298) ≤ 17.778175	deg 2	s(298) ≥ 17.270509
299	★ =	s(299) ≤ 17.822876	deg 2	s(299) ≥ 17.300297
300	★ =	s(300) ≤ 17.824124	deg 40	s(300) ≥ 17.330032
301	★ ≈	s(301) ≤ 17.846668		s(301) ≥ 17.359715
302	★ =	s(302) ≤ 17.881307		s(302) ≥ 17.389345
303	★ =	s(303) ≤ 17.920313		s(303) ≥ 17.418924
304	★ ≈	s(304) ≤ 17.934651		s(304) ≥ 17.448451
305	★ ≈	s(305) ≤ 17.952960		s(305) ≥ 17.477926
306	★ ≈	s(306) ≤ 17.963439		s(306) ≥ 17.507351
307	★ ≈	s(307) ≤ 17.981031		s(307) ≥ 17.536725
308	★ =	s(308) ≤ 18		s(308) ≥ 17.566048
309	★ =	s(309) ≤ 18		s(309) ≥ 17.595320
310	★ =	s(310) ≤ 18		s(310) ≥ 17.624543
311	★ =	s(311) ≤ 18		s(311) ≥ 17.653716
312	★ =	s(312) ≤ 18		s(312) ≥ 17.682840
313	★ =	s(313) ≤ 18		s(313) ≥ 17.711914
314	★ =	s(314) ≤ 18		s(314) ≥ 17.740939
315	★ =	s(315) ≤ 18		s(315) ≥ 17.769916
316	★ =	s(316) ≤ 18		s(316) ≥ 17.798843
317	★ =	s(317) ≤ 18		s(317) ≥ 17.827723
318	★ =	s(318) ≤ 18		s(318) ≥ 17.856554
319	★ =	s(319) ≤ 18		s(319) ≥ 17.885338
320	★ =	s(320) ≤ 18		s(320) ≥ 17.914074
321	★ O =	s(321) = 18		
322	★ O =	s(322) = 18		
323	★ O =	s(323) = 18		
324	O = R	s(324) = 18		